MEASUREMENT OF VISCOSITY USING A FALLING BALL VISCOMETER

Objectives:

To measure the viscosity of water/ methanol/glycerine as a function of temperature.

Apparatus:

(i) Falling ball viscometer (ii) Circulating water bath with an immersion thermostat,(iii) Viscometer balls, (iv) Pyknometer, (v) Retort stand, (vi) Weighing balance, (vii) Volumetric flask, (viii) Stopwatch, (ix) Thermometers, (x) Glasswares, (xi) Cleaning accessories and (xii) Distilled water/Methanol (laboratory grade)/Glycerine.

Introduction:

Viscosity is a measure of the resistance of a fluid which is being deformed by either shear stress or tensile stress. In everyday terms (and for fluids only), viscosity is "thickness". Thus, water is "thin", having a lower viscosity, while honey is "thick", having a higher viscosity. Put simply, the less viscous the fluid is, the greater its ease of movement (fluidity).In general, in any flow, layers move at different velocities and the fluid's viscosity arises from the shear stress between the layers that ultimately opposes any applied force.

In a Newtonian fluid, the relation between the shear stress and the strain rate is linear with the constant of proportionality defined as the viscosity. In the case of a non-Newtonian fluid, the flow properties cannot be described by a single constant viscosity. Some non-Newtonian fluids thicken when a shear stress is applied, e.g. cornflour suspensions, and some can become runnier under shear stress, e.g. non-drip paint. Industrially, understanding the viscous properties of liquids is extremely important and relevant to the transport of fluids as well as to the development and performance of paints, lubricants and food-stuffs.

Theoretical background

A body moving in a fluid is acted on by a frictional force in the opposite direction to its direction of travel. The magnitude of this force depends on the geometry of the body, its velocity, and the internal friction of the fluid. A measure for the internal friction is given by the dynamic viscosity η. For a sphere of radius r moving at velocity ω in an infinitely extended fluid of dynamic viscosity η, *G.G. Stokes* derived an expression for the frictional force:

$$F_1 = 6\pi \cdot \eta \cdot r \cdot \omega \qquad (1)$$

If the sphere falls vertically in the fluid, after a time, it will move at a constant velocity ω, and all the forces acting on the sphere will be in equilibrium (Fig. 1): the frictional force F_1 which acts upwards, the buoyancy force F_2 which also acts upwards and the downward acting gravitational force F_3. The two forces F_2 and F_3 are given by:

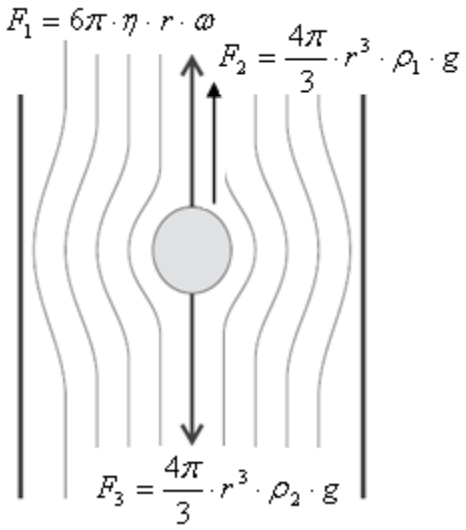


Fig. 1

$$F_2 = \frac{4\pi}{3} \cdot r^3 \cdot \rho_1 \cdot g \quad (2)$$

$$F_3 = \frac{4\pi}{3} \cdot r^3 \cdot \rho_2 \cdot g \quad (3)$$

where ρ_1 = density of the fluid

ρ_2 = density of the sphere

g = gravitational acceleration

The equilibrium between these three forces can be described by:

$$F_1 + F_2 = F_3 \quad (4)$$

The viscosity can, therefore, be determined by measuring the rate of fall ω:

$$\eta = \frac{2}{9} \cdot r^2 \cdot \frac{(\rho_2 - \rho_1) \cdot g}{\omega} \quad (5)$$

where ω can be determined by measuring the fall time t over a given distance s. The viscosity then becomes:

$$\eta = \frac{2}{9} \cdot r^2 \cdot \frac{(\rho_2 - \rho_1) \cdot g \cdot t}{s} = K \cdot (\rho_2 - \rho_1) \cdot t \quad (6)$$

Here $K = \frac{2}{9} \cdot r^2 \cdot \frac{g}{s}$, is known as ball constant. For the present experimental set up the ball constants are pre-determined using standard liquids with known value of viscosity.

In practice, equation 6 has to be corrected since the assumption that the fluid extends infinitely in all directions is unrealistic and the velocity distribution of the fluid particles relative to the surface of the sphere is affected by the finite dimensions of the fluid. If the finite inner radius R and length L of the cylinder containing the fluid is taken into account, the expression for viscosity is modified using Ladenburg correction as follows:

$$\eta = \frac{K \cdot (\rho_2 - \rho_1) \cdot t}{\left(1 + \frac{2.4\,r}{R}\right) \cdot \left(1 + \frac{3.3\,r}{L}\right)} \quad (7)$$

While Stokes' Law is straight forward, it is subject to some limitations. Specifically, this relationship is valid only for 'laminar' flow. Laminar flow is defined as a condition where fluid particles move along in smooth paths in lamina (fluid layers gliding over one another). The alternate flow condition is termed 'turbulent' flow. This latter condition is characterized by fluid particles that move in random in irregular paths causing an exchange of momentum between particles.

Units: The SI physical unit of viscosity is Pascal-second (Pa.s, equivalent to N.s/m^2 or kg/ms). The CGS unit is poise (P), named after Jean Louis Marie Poiseuille. It is more commonly expressed as centipoise (cP) or milli Pascal-second (mPa.s). The convertion factor is 1cP = 1mPa·s = 0.001Pa·s. Water at 20^{0}C has a viscosity of 1.0020 cP or 0.001002 kilogram/meter second.

Temperature dependence: The viscosity is a function of the temperature and the structure of the system. For many liquids the relation of the viscosity with temperature is described by an empirically determined exponential function

$$\frac{1}{\eta} = \varphi = C.e^{-\frac{E}{RT}} \qquad (9)$$

Here, φ is the fluidity of the liquid, C represents a system-dependent constant; R = 8.31441 $J \cdot K^{-1} \cdot mol^{-1}$, universal gas constant, T is the absolute temperature, E is an expression of the molar energy which is required to overcome the internal friction. This activation energy can be determined from the slope of the graph between ln η and 1/T, using the linear relation

$$\ln \eta = \left(\frac{E}{R}\right).\left(\frac{1}{T}\right) - \ln C \qquad (10)$$

Experimental Setup

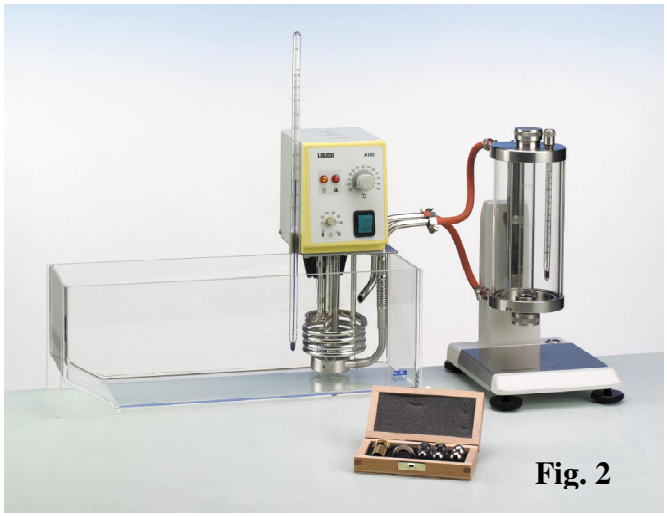


Fig. 2

1. Falling ball viscometer

The falling ball viscometer is based on Stokes' Law, and is what we will use in this laboratory investigation (Fig. 2). This type of viscometer consists of a circular cylinder of **inner diameter 15.94 ± 0.01 mm**, filled by the substance under investigation.

A standard ball is allowed to fall down this tube over a calibrated distance of 100 mm. The falling time is recorded and then utilized to determine the viscosity at a preset temperature. The measuring tube is jacketed by means of an outer glass tube, which encloses some room to be filled with a temperature controlled liquid. The measuring tube is fastened to the stand in such a way that its axis is inclined with respect to the vertical by 10° during the measurement to ensure that the ball falls along the axis of the tube. The measuring tube together with the jacket may be pivoted in order to turn the tube upside down again to let the ball return to the initial position before a measurement. The measuring tube is closed on both sides by two stoppers, one of which contains a capillary and a small reservoir. This stopper prevents undesirable changes of pressure in the liquid sample and has a passage for air bubbles when the temperature is being changed. The stand may be leveled by means of its water level and the leveling screws. The easily interchangeable thermometer allows a precise temperature control.

2. Temperature control

The Falling Ball Viscometer may be controlled in the temperature range 10 to 80ºC using the circulator provided. The sample should rest for at least 15 minutes in the measuring tube at the test temperature before the measurement is started.

Procedure:

1. Loading the sample:

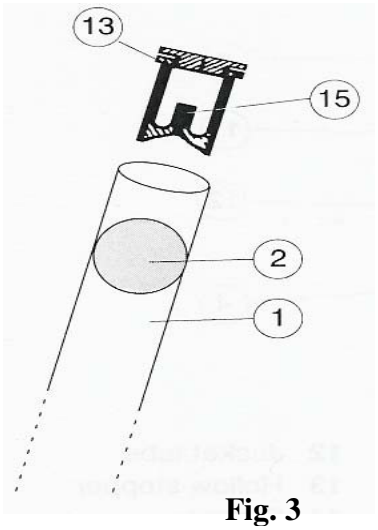


Fig. 3

All parts of the viscometer in direct contact with the sample must be kept clean and dry.
A sample volume of approximately 45 cm^3 is poured into the measuring tube **1** up to 20 mm below the tube (Fig. 3). The ball **2** is then introduced into the tube and the hollow stopper **13** placed on top. The liquid should reach a level just beyond the top of the capillary **15** in the stopper; top up if necessary. The sample in the tube must be free of air bubbles.
Before measurements are recorded, the ball should run up and down the tube at least once to improve the homogeneity of the samples and its temperature uniformity.

2. Setting the temperature:

To start with, set the temperature of the thermostat at room temperature. The liquid in the tube may take approximately 15 minutes to settle down at the set temperature.

3. Determination of density of the liquid:

First ensure that the thermometer attached to the outside circulating bath displays a steady temperature at the set value. Take a clean and dry pyknometer and weigh it empty using the weighing balance. Fill it with the sample liquid and fix it to the retort stand. Place it in the thermostatic water bath for approximately 15 minutes. Subsequent to bubble-free closure with the accompanying stopper and a quick external drying reweigh the filled pyknometer. Let m_1 and m_2 be the mass of the empty and filled pyknometer, respectively. Then the density of the liquid at a given temperature is determined as

$$\rho_{1=}\frac{(m_2-m_1)}{Volume\ of\ pyknometer\ (25\ cc)}$$

4. Measuring the falling times:

(a) The measuring tube assembly locates into a fixed position of 10º to the vertical. By turning the assembly over through 180º, the ball is set to the measuring position.
(b) Use a stopwatch to measure the time taken for the ball to fall between the lines marked at a distance of 10 cm apart on the measuring tube. The measuring period starts and ends when the lower edge of the ball crosses the upper and lower lines, respectively.
(c) Turning the assembly through 180º again returns the ball to its starting position. Take 3-5 readings for the falling time and calculate the average.

5. Temperature dependence: Repeat the steps from 2-4 for at least 5 different temperatures. Tabulate the data.

Observations:

Specification of Ball#1

Material	Density ρ_2(g/cm$^{3)}$	Diameter (mm)	Ball Const. (K)* mPa · s · cm^3/g · s	Recomm. visc. Range mPa.s
boron silica glass	2.2	15.81 ± 0.01	0.007	0.6-10

*The ball#1 is calibrated by the manufacturer and the ball const K is determined for a standard liquid of viscosity $\eta_{standard}$ = 4.63mPa.s and density $\rho_{standard}$ = 0.81g/cm^3 with average falling time t = 417.7s and the following equation is used.

$$K = \frac{\eta_{standard}}{(\rho_{standard} - \rho_1).t}$$

***Specification of* Viscometer tube**

Inner diameter of the measuring tube = 1.594 ± 0.001 cm,
Length of the cylinder = 22.7 ± 0.1 cm

Specification of Pyknometer Mass of empty pyknometer, m_1 = …..gm
Volume capacity = 25cc

Table:

Obs. No.	Temperature T (^{0}K)	Density ρ_1 (g/cm^3)	Falling time t_1(s)	t_2(s)	t_3 (s)	Average t (s)	Viscosity η (mPa.s)
1							
2							
..							

Graph: Plot a graph between ln η and 1/T to find the temperature dependence of viscosity of distilled water/methanol. Estimate the values of E and C from graph.

Discussions:

Precautions:

1. Make all the components clean and dry before starting measurement.
2. Avoid any physical contact with the circulating bath.
3. Avoid contaminating the balls, use tweezers or tissue paper to hold the balls.
4. The measuring cylinder should be bubble-free.
5. Measure the falling times carefully.

References:

1. D.R. Rohendra etal., Europian J. Phys. Vol. 33, pp. 1457, (2012)
2. Thermo haake-Falling Ball Viscometer (user manual) for calibration of viscometer.